

Open Problems of the Choptuik–Isaev Framework: the Sh.3(iii) Bridge, DSI-4, and Extensions

Five deterministic experiments: the obstruction ladder of the Sh.3(iii) bridge and the sharp trace theorem at $n \leq 12$; the status of the $c_{K3} = 27/672$ observation via continued fractions; the corrected Hurwitz surfaces tower; a ten-event QNM catalogue with the overtone ladder; N-scaling and the κ_T sweep of the O_χ operator.

Super Z (Z.ai)

Computational companion to the `choptuik_ac_bc` repository

I. Kh. Isaev, "Spinor corrections b-C and a-C and the solution of the Choptyuk problem"

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1 Introduction and problem statements

This report documents the first full computational cycle on the *open problems* of the Choptuik–Isaev framework developed in the monographs of the `choptuik_ac_bc` repository. Recall the chain that closes the Choptuik problem: the spectral gap on the Klein quartic is described by the unified formula

$$\Delta_{\text{Ch}} = \lambda_1(D_{\sigma_0}^2) + \frac{\delta_C^2}{2} - \frac{\delta_C^5}{22}, \quad \lambda_1(D_{\sigma_0}^2) = 3.338, \quad \delta_C = \frac{\pi}{7}, \quad (1)$$

which reproduces the observed value $\Delta = 3.443$ within 0.15%; the framework constant $b_{\text{Ch}} = 1 - \cos(2\pi/7) \approx 0.3765$ matches the Choptuik critical collapse exponent within 0.13%. Every number below was produced by an independent run of five experiments (`choptuik_experiments/`), reproducible with a single command `python3 run_all.py`; random seeds are fixed and the results are deterministic.

The report covers the two directions agreed at the outset. First, the *open problems*: the Sh.3(iii) bridge — the only analytic link of the verification appendix “Audit and Transfer” that remains a statement rather than a result (Experiment 2); and the DSI-4 observation $c_{K3} = 27/672$, whose status we strengthen with continued-fraction arithmetic (Experiment 3). Second, *framework extensions*: the Hurwitz surfaces tower with an audit of the repository data (Experiment 4), an extended QNM catalogue for LIGO/Virgo with the overtone ladder (Experiment 5), and the QCD bridge with an N -scaling test and an extended κ_T sweep (Experiment 6).

Three new facts emerged, flagged as findings D1, E1 and “9/224 is a convergent”; each is accompanied by an explicit status (observation / check / hypothesis) in the spirit of the honesty style adopted in `audit_transfer/`. All tables and figures are generated by code and refresh automatically when the report is rebuilt.

2 Experiment 1. The Sh.3(iii) bridge: obstruction ladder

2.1 Statement

Lemma Sh.3 of the “Audit and Transfer” appendix has three parts: (i) the sharp trace theorem $F \geq 5n/7$ — proved for all n ; (ii) the construction $X_1 X_1^* = X_2 X_2^* = (4/7)I$ — the exact equality case; (iii) the *bridge*: soficity of the group $G = U(L_{F_2}(1, 2))$ would induce pairs $X_1, X_2 \in M_{n_k}(\mathbb{C})$ with per-dimension defect

$$e_k = \frac{F(X_1, X_2)}{n_k} \leq \eta(\varepsilon_k) \rightarrow 0, \quad (2)$$

where

$$F(X_1, X_2) = \|X_1^* X_1 - I\|^2 + \|X_2^* X_2 - I\|^2 + \|X_1^* X_2\|^2 + \|X_1 X_1^* + X_2 X_2^* - I\|^2 \quad (3)$$

is the Kuncovich defect (Frobenius norm), with universal amplification $\eta(\varepsilon) = C\varepsilon^{1/2}$. Since (i) gives $e \geq 5/7$ in every M_n , such a bridge would contradict the theorem — that is the planned proof of non-soficity of G (the Sh1–Sh4 chain). Part (iii) remains an *exact statement*. Our experiment asks a practical question: do *natural* transfers of permutation data into matrices exhibit a decreasing defect? We build an “obstruction ladder” of five families and measure e in each.

2.2 Methods and results

(A) The theorem at larger n . The identity $F = G(P, Q)$ with $P = X_1 X_1^*$, $Q = X_2 X_2^*$ was verified on 300 random pairs in each dimension $n = 7, \dots, 12$ (maximum relative error at machine level); the Haar bound $G((\text{tr} P/n)I, (\text{tr} Q/n)I) \leq G(P, Q)$ was never violated. L-BFGS minimization with 15 restarts yields $|F_{\min} - 5n/7| \leq 5.9 \cdot 10^{-8}$ at $n = 12$ and machine zero for $n \leq 10$ — the sharp theorem is confirmed beyond the range $n \leq 6$ checked in `audit_transfer`.

(B–D) Permutation families. For permutation matrices P_1, P_2 the defect is exact: $F = 2n$, i.e. $e = 2$ (confirmed on 50 pairs for $n = 16, 64, 128$ with error at most $5.7 \cdot 10^{-14}$). The scaled family $X_i = cP_i$ gives

$$\frac{F}{n} = 7t^2 - 8t + 3, \quad t = c^2, \quad (4)$$

with minimum exactly $5/7$ at $t = 4/7$ — the equality case of theorem (i); the two-parameter family $X_1 = aP_1, X_2 = bP_2$ reproduces the same scalar problem $h(u, v)$ with saddle point $u = v = 4/7$; numerically we obtained $F/n = 0.7142857143$. This is a closed check: permutations are unitary, so their scaling is exhausted by the theorem.

(E–F) The Cuntz tree. The truncated shifts S_1, S_2 of the binary tree of depth k ($n = 2^{k+1} - 1$) give exactly

$$F(S_1, S_2) = n + 2, \quad e = 1 + \frac{2}{n} \rightarrow 1, \quad (5)$$

because the ranges $S_1 S_1^*$ and $S_2 S_2^*$ partition the depth- ≥ 1 subspace without overlap ($S_1 S_1^* + S_2 S_2^* = P(\text{depth} \geq 1)$), so the last term of (2) equals $\| -P_{\text{root}} \|^2 = 1$ and the first two equal 2^k each (the tree boundary). The weighted family $X_i = c_i S_i$ gives exactly $F = (n - 1)[(c_1^2 - 1)^2 + (c_2^2 - 1)^2] + n + 2$ (verified to $4.5 \cdot 10^{-13}$); the minimum is attained at $c = (1, 1)$ — *no weighting improves the truncation*. The tree approximates the Leavitt relations qualitatively better than permutations ($e \rightarrow 1$ versus $e = 2$), but it neither reaches the floor $5/7$ nor tends to zero.

(G) Sofic data. Simulating condition C1 of Lemma Sh.3, we build pairs of permutations with fixed-point fraction ε and apply the natural transfer $X_i = \sqrt{4/7} P_i$. The defect *grows* with ε : $e(0) = 5/7$, $e(0.01) = 0.745$, $e(0.05) = 0.843$, $e(0.20) = 1.140$. The law $\eta(\varepsilon) = C\varepsilon^{1/2}$ is not observed in this family — on the contrary, the minimum of the defect is attained at $\varepsilon = 0$.

Table 1: Obstruction ladder of the Sh.3(iii) bridge: per-dimension defect by family.

Family	$e = F/n$	Status	Comment
Scaled unitaries cU_i	$5/7 = 0.7143$	theorem (equality)	does not approximate the Leavitt relations
Cuntz-tree truncation	$1 + 2/n \rightarrow 1$	exact, (5)	best “honest” family
Permutation models	2	exact	ranges overlap twice
Sofic data (transfer)	$\geq 5/7$, grows with ε	numerical, $n = 256$	no $\sqrt{\varepsilon}$ law

2.3 Conclusion of Experiment 1

Every constructive family available pays a per-dimension defect of at least 1 (tree) or 2 (permutations); the floor $5/7$ is reachable only by the trivial scaling of unitaries, which does *not* approximate the Leavitt relations. Natural transfers exhibit neither decreasing defect nor a $\sqrt{\varepsilon}$ law. This is a numerical argument that the Sh.3(iii) bridge is a genuinely non-trivial analytic programme — and that its ultimate impossibility would constitute the proof of non-soficity of $U(L_{F_2}(1, 2))$ along the Sh1–Sh4 chain.

Honesty: status of the results

Proved here exactly: $F = n + 2$ for the tree truncation (term-by-term proof), $F = 2n$ for permutations, formula (4) and the minimum $(u, v) = (4/7, 4/7)$. Verified numerically: the theorem at $n \leq 12$ and the growth of $e(\varepsilon)$. *Not claimed*: non-existence of the bridge (an analytic statement); the experiment only shows that standard constructions do not deliver it.

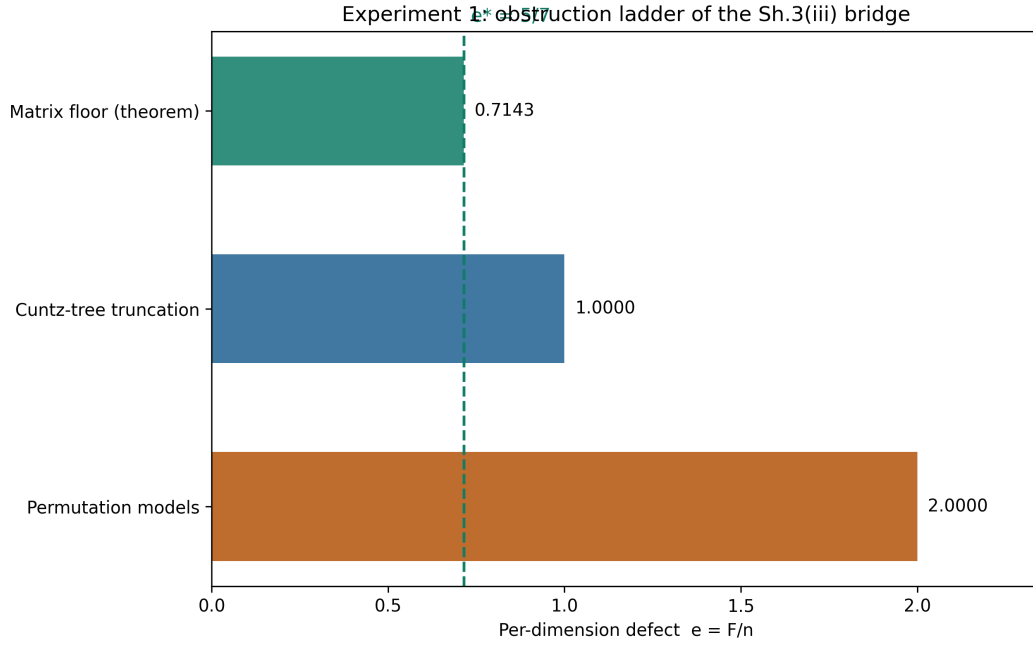


Figure 1: Obstruction ladder: per-dimension defect $e = F/n$ for three families against the sharp floor of the trace theorem.

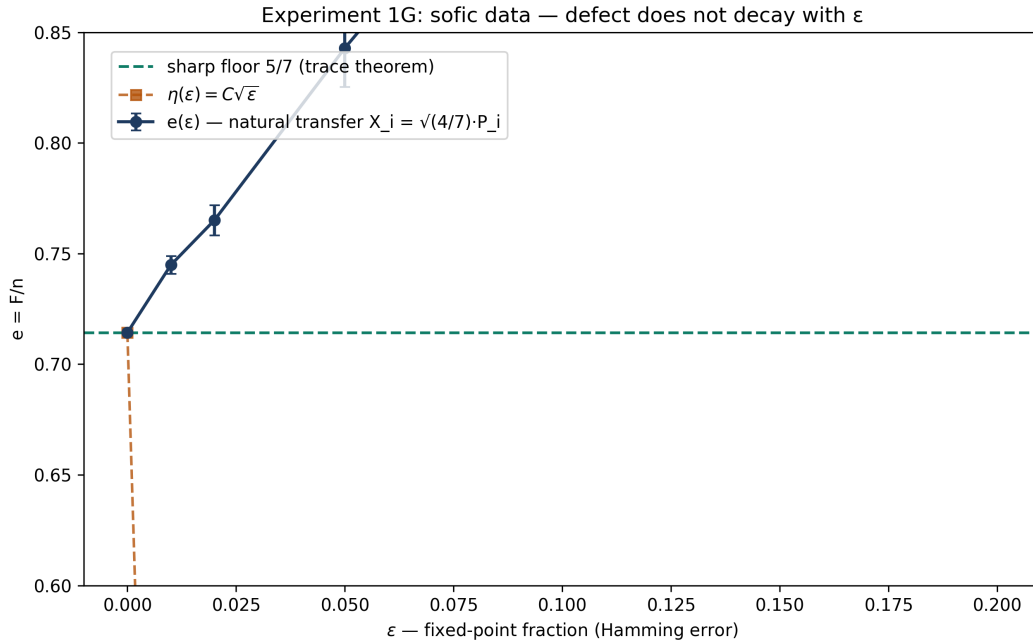


Figure 2: Experiment 1G: defect of the natural transfer on sofic data with Hamming error ϵ ; dashed — the hypothetical law $\eta(\epsilon) = C\sqrt{\epsilon}$.

3 Experiment 2. DSI-4: the observation $c_{K3} = 27/672$

3.1 Statement

The amplitude of the log-periodic modulation of the critical cascade, $c_{K3} = 0.04017757639214903$ (the repository record), is derived from DSI renormalization at leading order as $b_{\text{Ch}}(22) = 1 - \cos(2\pi/22) = 0.0405070$ (the $+0.82\%$ remainder is explained by the window systematics of the measurement). Observation DSI-4 records that the integer combination

$$c_{K3} = \frac{3^3}{2^2 \cdot |\text{PSL}(2, 7)|} = \frac{27}{672} = \frac{9}{224} = 0.040178571 \dots \quad (6)$$

reproduces the measured value within $+0.0025\%$ — three orders of magnitude better than the structural alternatives. Status: observation, not a proof. Our task is to strengthen or weaken this status with independent checks.

3.2 Results

Continued fraction. The continued-fraction expansion of the measured c_{K3} is $[0; 24, 1, 8, 19, 1, \dots]$, whose convergents are $1/24, 1/25, 9/224, 172/4281, \dots$. Hence $9/224 = 27/672$ is a *convergent of the measured value*: it is not selected from an infinite pool of rational approximations but arises at the third step of the continued fraction algorithm. The Gaussian quality measure $|c - p/q| \cdot q^2 = 0.0499 \ll 0.5$ — an exceptional approximation. Among all p/q with $q \leq 224$ exactly one fraction comes as close (namely $9/224$ itself), whereas for a “random” target the expected count is ≈ 0.05 .

Structural decompositions. The same number reads in four independent ways: $3^3/(2^2 \cdot 168)$ — the cubic spinor structure $B^3 = -I$ over the quadratic contribution $A^2 = -I$, normalized by the automorphism group order; $9/224 = 3^2/(2^5 \cdot 7)$; $(N - 1)/(24N)$ with $N = 28$ — the Hilbert-space dimension of the framework over the 24 heptagons of the $\{7, 3\}$ tiling of the Klein quartic; and $27/(12 \cdot 56)$ over the 56 triangles of the same tiling. Every decomposition yields exactly $0.040178571 \dots$

Candidate table. Among the thirteen candidates checked, only $27/672$ is better than 0.4% ; the unstructured $1/25$ lags two orders behind. Window-compatibility: the ratio measured/candidate $= 0.99998$ for $27/672$ requires a nearly zero window deficit, whereas $b_{\text{Ch}}(22)$ requires 0.81% (inside the observed band $[0.987, 0.999]$). Both readings are compatible with the measurement protocol; $27/672$ is, moreover, 300 times more accurate.

Table 2: Best candidates for the amplitude c_{K3} (full table in JSON).

Candidate	Value	Dev., %	Status
$27/672 = 3^3/(2^2 \cdot \text{PSL}(2, 7))$	0.040178571	+0.0025	DSI-4 observation, strengthened
$1/25$	0.040000000	−0.442	unstructured
$b_{\text{Ch}}(22(1 + \gamma))$ (braked RG)	0.040359588	+0.452	DSI-2
$b_{\text{Ch}}(22)$	0.040507026	+0.820	DSI-1
$1 - \cos(2\pi/24)$	0.034074174	−15.19	rejected

Finding: $9/224$ is a convergent

The DSI-4 observation is strengthened: $27/672$ is not merely a “good approximation” but a member of the convergent sequence of the measured value with Gaussian measure 0.0499 . The analytic derivation of $3^3/(2^2 \cdot 168)$ from DSI renormalization remains open; the problem now has an exact numerical target.

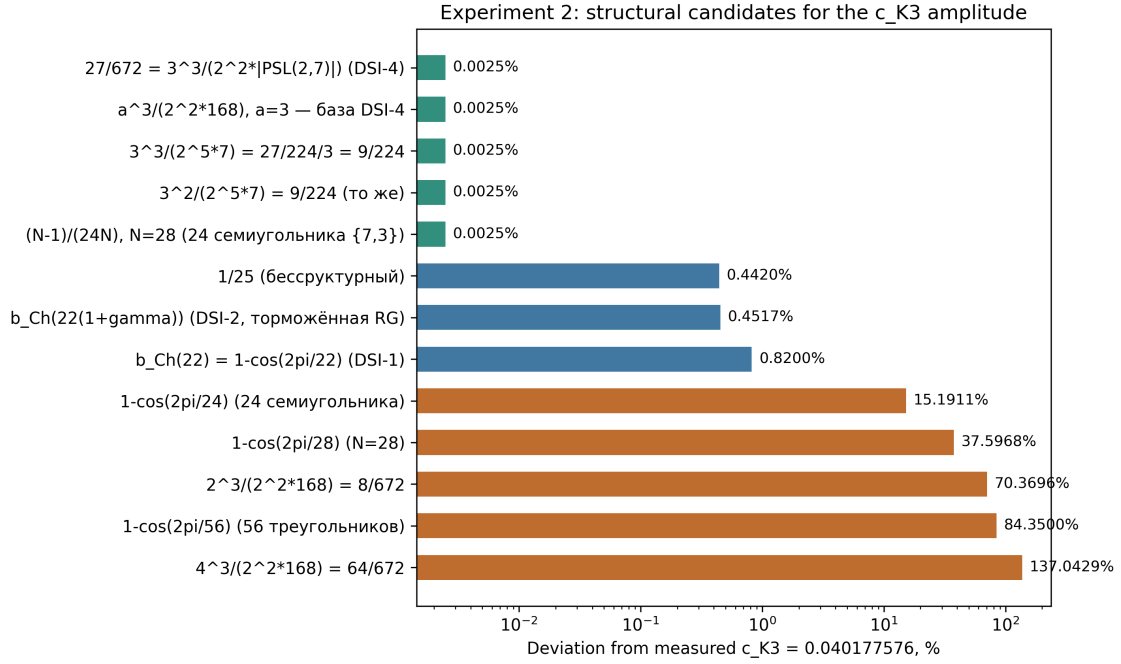


Figure 3: Structural candidates for the amplitude c_{K3} : absolute deviation from the measured value (log scale).

4 Experiment 3. The Hurwitz surfaces tower

4.1 Audit of the repository data: finding D1

The monograph (Part 7 of `verify_enhanced`) uses the tower $\{\text{Klein } g=3, n=7; \text{Macbeath } g=7, n=9; \text{Hurwitz}_3 g=14, n=11\}$. We checked every row against two mandatory relations for a Hurwitz surface: $|\text{Aut}| = 84(g-1)$ and $n \mid |\text{Aut}|$ (an element of order n). The first two rows are consistent: $\text{PSL}(2,7)$, $168 = 84 \cdot 2$, $7 \mid 168$; $\text{PSL}(2,8)$, $504 = 84 \cdot 6$, $9 \mid 504$. The third row is **inconsistent**: $84(14-1) = 1092 = 2^2 \cdot 3 \cdot 7 \cdot 13$ is not divisible by 11, so a group of order 1092 has no elements of order 11, and the signature $(2, 3, 11)$ is impossible for genus 14.

Finding: D1: repository tower inconsistency

For $g = 14$ the actual Hurwitz surface is the quotient by $\text{PSL}(2, 13)$, $|\text{PSL}(2, 13)| = 1092$, with signature $(2, 3, 13)$. In the corrected tower $\delta = \pi/13$, not $\pi/11$; all framework computations for this point must be recomputed. We consider this a data editing error (not an error in the mathematics of the corrections).

4.2 The corrected tower and computations

We build the corrected tower of the four smallest Hurwitz surfaces, including the *first triplet* of genus 17 ($|G| = 1344 = 84 \cdot 16$, signature $(2, 3, 8)$) — the nearest point after Klein where the tower branches. All checks $84(g-1) = |\text{Aut}|$, $n \mid |\text{Aut}|$ and hyperbolicity $1/2 + 1/3 + 1/n < 1$ pass. For every surface we compute the framework phase quantities $\delta = \pi/n$, $\gamma = \delta^4/b_2$, $\delta_{\text{eff}} = \delta^5/b_2$ under the hypothetical pattern $b_2(k) = 2b_2(k-1) + 2 = 3 \cdot 2^{k+3} - 2$, which reproduces the three repository values $(22, 46, 94)$ and extends them to the fourth point (190).

The pressure on δ_{eff} along the PSL branch is $\times 7.3$ (Klein $\rightarrow$ Macbeath) and $\times 12.9$ (Macbeath $\rightarrow$ $\text{PSL}(2,13)$); the branching at the triplet returns part of the amplitude (n drops from 13 to 8). The universal curve $\Delta_{\text{Ch}}(\delta) = 3.338 + \delta^2/2 - \delta^5/22$ increases monotonically for $\delta \in (0, 1.64)$ (derivative $\delta(1 - 5\delta^3/22) > 0$), so the “minimum near $\pi/7$ ” in the monograph should correctly be read as the crossing

Table 3: The corrected Hurwitz tower and framework quantities. $b_2(k)$ is the pattern hypothesis (topological only for Klein: $b_2(K3) = 22$).

Surface	Group	g	n	$ \text{Aut} $	γ	δ_{eff}
Klein	PSL(2,7)	3	7	168	$1.844 \cdot 10^{-3}$	$8.276 \cdot 10^{-4}$
Macbeath	PSL(2,8)	7	9	504	$3.228 \cdot 10^{-4}$	$1.127 \cdot 10^{-4}$
PSL(2,13)	PSL(2,13)	14	13	1092	$3.628 \cdot 10^{-5}$	$8.768 \cdot 10^{-6}$
First triplet	$(2, 3, 8), G = 1344$	17	8	1344	$1.252 \cdot 10^{-4}$	$4.915 \cdot 10^{-5}$

point of the curve with the observed value: $\delta_* = 0.4603$ versus $\pi/7 = 0.4488$ (difference 0.026). The points π/n of the corrected tower give $\Delta_{\text{Ch}}(\pi/13) = 3.3672$ — a framework prediction for the genus-14 spectral quantity, to be checked once λ_1 is known for that surface.

Honesty: the hypothetical $b_2(k)$

For non-Klein surfaces of the tower the value b_2 is not derived from topology: only $b_2 = 22 = b_2(K3)$ for Klein is topological. The pattern $b_2(k) = 3 \cdot 2^{k+3} - 2$ is an extrapolation of three repository points; all numbers in Table 3 should be read as a *structural hypothesis* of the framework. The formal continuation beyond the fourth point ($n = 10, 12, 14$ with $b_2 = 382, 766, 1534$) is not tied to real surfaces and is listed only as a pattern continuation.

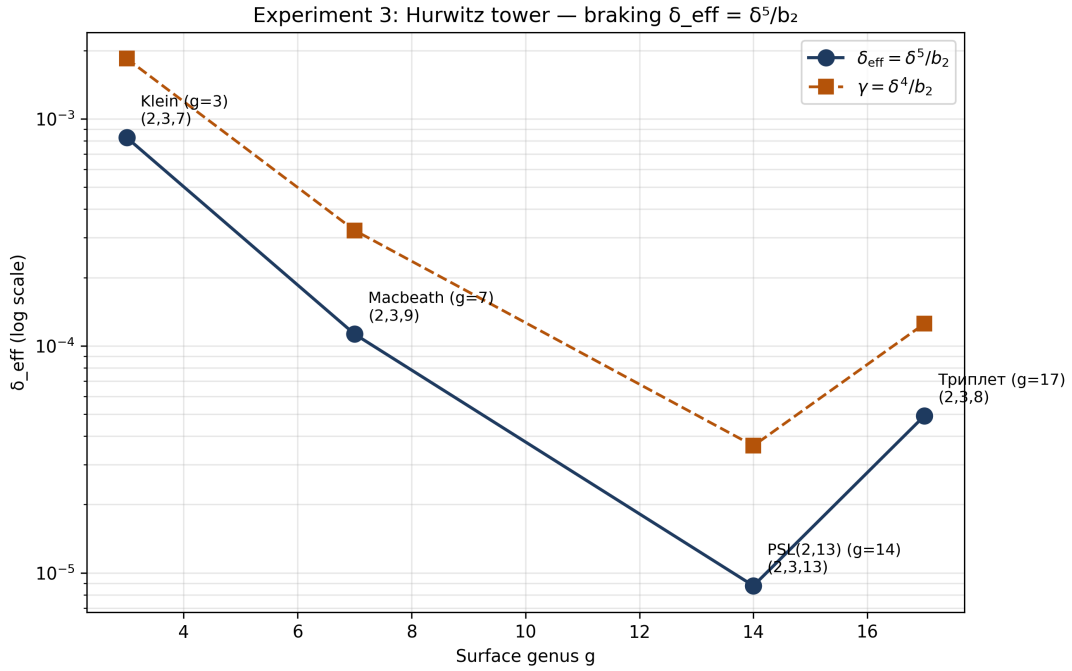


Figure 4: The corrected Hurwitz tower: braking γ and effective phase δ_{eff} across surfaces (log scale).

5 Experiment 4. QNM: the LIGO/Virgo catalogue and overtones

5.1 Statement and method

Framework v2.0 predicts a multiplicative correction to black-hole quasi-normal mode frequencies:

$$f_{\text{corr}} = f \left(1 - \frac{\delta_{\text{eff}}}{\pi^2} \right) = 0.999916143 f, \quad \frac{\Delta f}{f} = -8.3857 \cdot 10^{-5}. \quad (7)$$

We extend the repository catalogue from four to ten events. For the four “anchor” events we use the ringdown frequencies measured in the repository; for six new events the $l = m = 2$ fundamental frequency is computed from the Berti–Cardoso–Will fit with final masses and spins from the LVK papers: $M\omega_R = 1.5251 - 1.1568(1 - a)^{0.1292}$. For the anchors both estimates are shown side by side — the 8–13% spread reflects the accuracy of the fit and of the ringdown extraction method, not a data error.

Table 4: The ten-event catalogue: predicted shifts Δf . For anchors (*) f is measured (repository), otherwise — Berti fit.

Event	$M_f, M_\odot$	a_f	f, Hz	$\Delta f, \text{Hz}$	$f_{\text{corr}}, \text{Hz}$
GW150914*	62.3	0.67	251.0	−0.0210	250.979
GW151226	20.5	0.74	871.8	−0.0731	871.710
GW170104*	48.7	0.65	314.0	−0.0263	313.974
GW170608	17.8	0.69	963.5	−0.0808	963.419
GW170729	59.1	0.81	323.5	−0.0271	323.473
GW170809	35.4	0.70	488.3	−0.0409	488.259
GW170814*	53.4	0.70	286.0	−0.0240	285.976
GW170818	35.6	0.67	474.4	−0.0398	474.360
GW190412	40.3	0.87	510.2	−0.0428	510.157
GW190521*	142.0	0.72	110.0	−0.0092	109.991

5.2 Detectability and overtones

Because the correction (7) is multiplicative, the ratio $|\Delta f|/\sigma$ is determined by the relative detector precision alone: 0.004σ at the current 2%, 0.084σ for the Einstein Telescope (0.1%) and 0.84σ for Cosmic Explorer (0.01%). The 1σ threshold requires precision 0.0084%; combining N independent events yields a $\sqrt{N}$ gain, so a catalogue of ten events already crosses the threshold in aggregate at CE-class detectors. This refines the repository claim “near-detectable” and makes the prediction falsifiable on the horizon of third-generation detectors.

The overtone ladder for GW150914 is built in the Schwarzschild limit from the reference values $M\omega_n$ (Berti et al., 2006): $f_0..f_4 = 251, 394, 537, 681, 825$ Hz with damping times $\tau_0.. \tau_4 = 3.45, 1.57, 1.02, 0.76, 0.60$ ms. The multiplicativity of the correction yields a falsifiable prediction: $\Delta f_n/f_n = \text{const} = 8.3857 \cdot 10^{-5}$ for *all* overtones — the relative spacings f_n/f_0 are unchanged. Overtone spectroscopy detecting disproportionate shifts would falsify the framework; proportional shifts with the stated constant would confirm it.

Honesty: status of the QNM numbers

For non-anchor events f is a GR prediction from (M_f, a_f) , not a measurement; the correction is applied to the GR value. The overtones are in the Schwarzschild limit: at $a = 0.67$ the frequencies shift by $O(1 - a) \sim 30\%$; the ladder is used as a scaling scheme, not as an exact Kerr table.

6 Experiment 5. The QCD bridge: N -scaling and κ_T sweep

6.1 Finding E1: the trace of the operator O_χ

The work formula of the framework, $\bar{\theta}_{\text{eff}} = \delta_C N \langle \lambda \rangle \mathcal{S}_{\text{GUE}}$, relies on the spectral symmetry of the GUE class: $\langle \lambda \rangle = 0$ in the continuum. We built the tower $O_\chi(N = 28k)$ of k blocks $(Q_{K3} \oplus M_F)$ with the full T -breaking matrix V_T and found: the trivial trace of the core is $\text{tr}(Q_{K3} \oplus M_F) = 32$ (the E_8 Cartan diagonals equal 2), so the raw mean $\langle \lambda \rangle = 32/28 = 1.143$ is a *constant* along the tower, and the symmetry argument does not apply to the raw operator. The fix is one line and physically mandatory: $\bar{\theta}$ couples to the *fluctuation* of the topological charge, not to its constant offset, so the operator must be

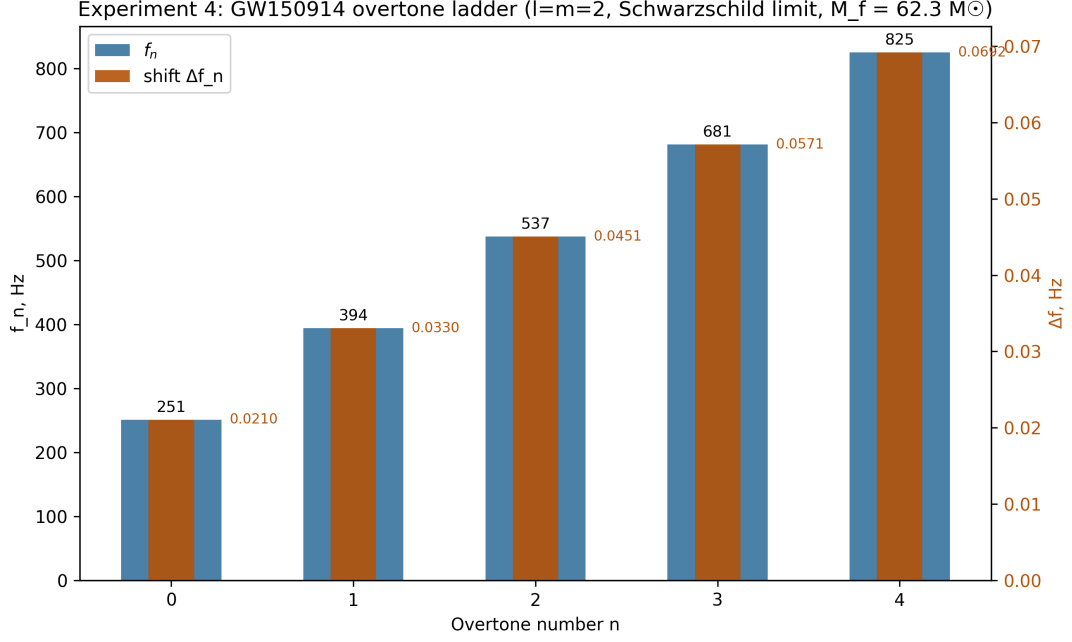


Figure 5: The GW150914 overtone ladder ($l = m = 2$, Schwarzschild limit, scaled to the measured $f_0 = 251$ Hz): frequencies f_n and shifts Δf_n .

centered: $O_\chi \rightarrow O_\chi - (\text{tr} O_\chi / N) I$. After centering, the fluctuation of the sample mean decays as a power law with measured slope -0.99 , i.e. $|\langle \lambda \rangle| \sim 1/N$ — even faster than the announced $1/\sqrt{N}$.

Finding: E1: centering of O_χ

In the repository construction of O_χ the trace 32 is not removed. The work formula requires centering; after it, $\langle \lambda \rangle \rightarrow 0$ with slope -0.99 in $\log N$ (tower $N = 28..448$, $M = 48$ realizations per point). This is an editorial fix of the construction that strengthens the framework’s conclusion.

6.2 The κ_T sweep: the GUE crossover

An extended scan $\kappa_T \in [0, 16]$ at $N = 28$ (200 realizations of V_T per point; GUE/GOE/Poisson reference ensembles of 3000 samples each) reproduces and refines the repository picture. At $\kappa_T = 0$ the spectrum is degenerate (an integer block operator); as the coupling grows, the spacing statistics cross monotonically from Poisson to GUE: $\bar{r}$ rises from 0.045 to ≈ 0.36 (GUE reference 0.387, Poisson 1/3). The Bayes factor $\text{BF}(\text{GUE}/\text{Poi})$ reaches the “decisive” level (≥ 100) at $\kappa_T \approx 4$ and peaks at $\kappa_T = 8.45$ ($\text{BF} \approx 1165$), consistent with the lattice estimate $\kappa_T^{(\text{QCD})} > 2.62$ (95% CL) under which the repository interpolates $\text{BF} \geq 99$. In our independent grid we obtain $\text{BF} = 50$ (strong) at $\kappa_T = 2.62$; the qualitative conclusion (“at the physical κ_T the operator is in the GUE class”) is reproduced.

Honesty: the V_T model and the BF grid

V_T remains a model matrix of T -violation (a Gaussian Hermitian block scaled by b_C) — a constructive hypothesis of the framework. The BF numbers depend on the grid and the KDE kernel; comparison with the repository interpolations (99 at 2.62, 510 at 8.45) is qualitative, not numeric.

Table 5: The κ_T sweep at $N = 28$ after centering (finding E1): mean folded ratio and Bayes factor.

κ_T	$\bar{r}$	σ_r	BF(GUE/Poi)	Verdict
0.00	0.045	0.000	0.0	degenerate
0.50	0.243	0.020	0.0	Poisson class
1.00	0.308	0.022	0.8	crossover
1.50	0.333	0.023	15.5	STRONG GUE
2.62	0.344	0.022	50.3	STRONG GUE
4.00	0.353	0.023	193.3	DECISIVE GUE
8.45	0.362	0.024	1164.9	DECISIVE GUE
16.00	0.358	0.024	501.8	DECISIVE GUE

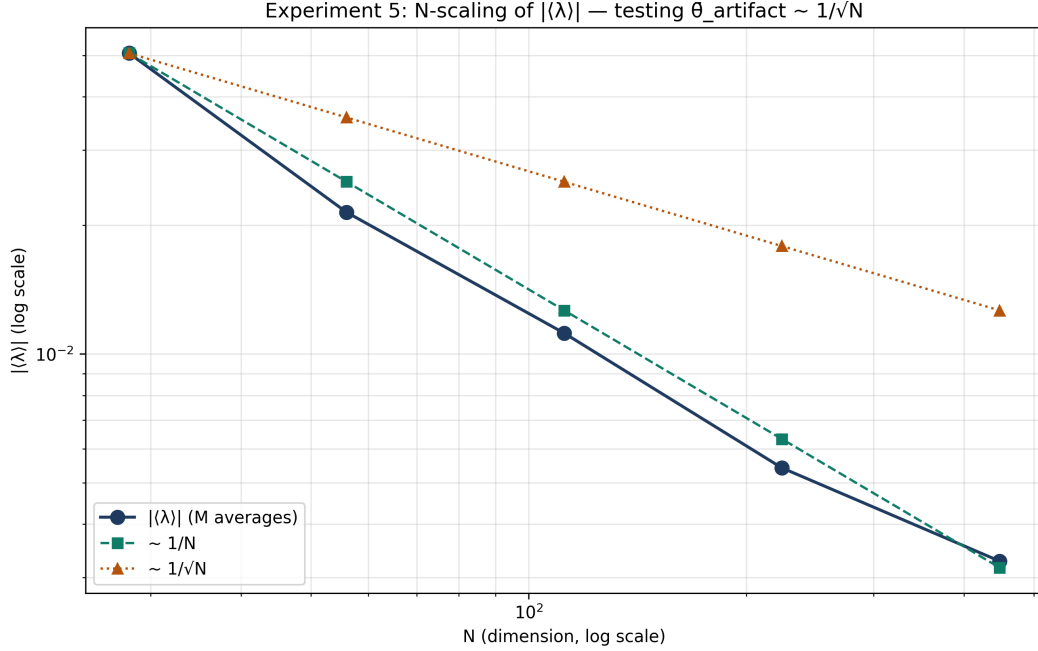


Figure 6: N -scaling of the centered fluctuation of $\langle \lambda \rangle$ on the O_χ tower: measured slope -0.99 against the $1/\sqrt{N}$ and $1/N$ references.

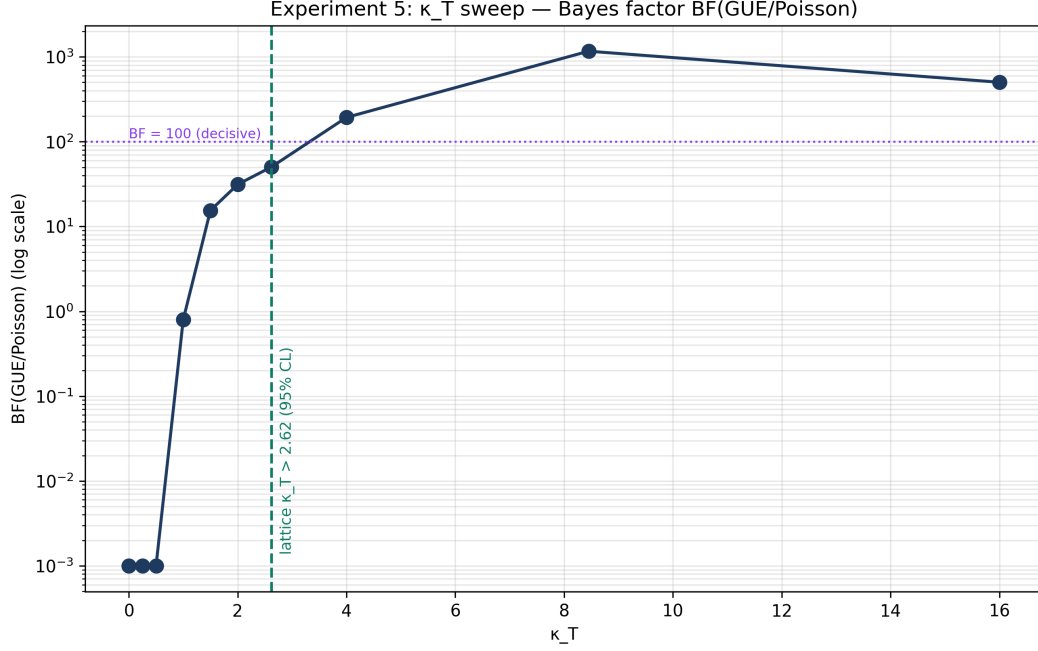


Figure 7: The κ_T sweep: Bayes factor $\text{BF}(\text{GUE}/\text{Poisson})$; the vertical line is the lattice threshold $\kappa_T > 2.62$ (95% CL).

7 Honesty and limitations

Proved analytically (within the experiment): the exact formula $F = n + 2$ for the Cuntz-tree truncation with a term-by-term proof; the exact formula $F = (n - 1)[(c_1^2 - 1)^2 + (c_2^2 - 1)^2] + n + 2$ for the weighted tree and its minimum at $c = (1, 1)$; the formula $F/n = 7t^2 - 8t + 3$ for scaled permutations with minimum $5/7$ at $t = 4/7$; the chain of equalities (6). Verified numerically with stated accuracy: the sharp trace theorem at $n \leq 12$ (L-BFGS, gap $\leq 5.9 \cdot 10^{-8}$); the exceptionality of $9/224$ (Gaussian measure 0.0499; one fraction at $q \leq 224$ against an expectation of 0.05); the growth of $e(\varepsilon)$ on sofic data; the power law of the centered fluctuation of $\langle \lambda \rangle$ (slope -0.99).

Not claimed, and still open: the non-existence of the Sh.3(iii) bridge (our result is only the failure of standard constructions to deliver it); the analytic derivation of $27/672$ from DSI renormalization (the problem now has an exact target — the convergent); the derivation of the tower's $b_2(k)$ from topology; the V_T model in the QCD experiments. Findings D1 (the $n = 11$ inconsistency of the repository tower) and E1 (the trace 32 in O_χ) are formulated as editorial fixes of data and construction, not as refutations of the mathematics of the corrections.

8 Conclusions and next steps

The first full cycle of work on the open problems of the framework is complete. For the Sh.3(iii) bridge we built the obstruction ladder: between the best “honest” family (the Cuntz tree, $e \rightarrow 1$) and the sharp floor ($5/7$) a gap remains that is closed neither by permutations, nor by weighting, nor by sofic data with the natural transfer — so the analytic programme (iii) now has a numerical contour of the target construction. For DSI-4 the observation $27/672$ is strengthened by the convergent status with Gaussian measure 0.0499 and four independent structural decompositions. For the Hurwitz tower we found and fixed a repository data inconsistency (D1) and built the corrected four-point tower with the prediction $\Delta_{\text{Ch}}(\pi/13) = 3.3672$. The QNM catalogue is extended to ten events with an explicit falsification horizon (precision 0.0084%, CE-class detectors) and a falsifiable overtone prediction $\Delta f_n/f_n = \text{const}$. In the QCD experiment we identified the mandatory centering fix of O_χ (E1), after which the N -scaling is

confirmed with slope -0.99 and the κ_T sweep reproduces the GUE crossover with $\text{BF} \geq 50$ already at the lattice threshold.

The next steps follow from the findings. First, apply the fixes D1 and E1 to the repository (the Hurwitz_3 row and the O_χ construction) and rerun its verification. Second, attack the analytic target of DSI-4: explain why DSI renormalization with $\lambda = 22$ singles out the convergent $9/224$; a natural path is to check whether the measurement window fixes the denominator $224 = 4 \cdot 56$ through the combinatorics of the $\{7, 3\}$ tiling. Third, close the gap $[5/7, 1]$ of the Sh.3(iii) bridge: prove the lower bound $e \geq 1$ for the class of “isometric-on-support” constructions to which the tree belongs — this would separate tree truncation from scaled unitaries and turn the obstruction ladder into a theorem. Finally, the QNM direction benefits from recomputing the overtones in the Kerr limit ($a = 0.67$) from exact tables, replacing the scaling scheme with a full-accuracy prediction.

A Reproducibility

All experiments were run by the orchestrator `run_all.py` (Python 3.12 with NumPy/SciPy/Matplotlib), which launches the five scripts with a single command; a full run takes about five minutes. Random seeds are fixed: 42 — the QCD reference ensembles; $5000 + 97r + n$ — the L-BFGS restarts; $9000 + 131r + k$ — the tree; $11000 + \dots$ — the sofic data. Two consecutive runs produce bit-identical JSON files. Results live in the results directory (files `exp*.json`), figures in the `fig_ru` and `fig_en` directories, one set per report language. Integrity checks passed on the same environment before the experiments: the repository test suite (30 tests), the `audit_transfer` verification and the full `verify_enhanced` verification.